Prayas JEE AIR (2025)

WEEKLY TEST 26

DURATION : 180 Minutes | **DATE : 04/01/2025** | **M.MARKS : 300**

Topics Covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 75 questions. The maximum marks are **300**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Mathematics)** and having **25** questions in each part in which first **20** questions are of Objective Type and Last **5** questions are integers type and all **25** questions are **compulsory.**
5. There is only one correct response for each question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number: _____

OMR Bar Code Number: _____

Candidate's Signature: _____ **Invigilator's Signature** _____

SECTION-I (PHYSICS)

Single Correct Type Questions

1. An object of mass 5 kg is thrown vertically upwards from the ground. The air resistance produces a constant retarding force of 10 N throughout the motion. The ratio of time of ascent to the time of descent will be equal to (Use g = 10 ms^{-2})

(1) 1 : 1 (2) $\sqrt{2}:\sqrt{3}$.

(3) $\sqrt{3}:\sqrt{2}$ (4) 2 : 3

2. 100 balls each of mass *m* moving with speed *v* simultaneously strike a wall normally and are reflected back with the same speed, in time *t* s. The total force exerted by the balls on the wall is

(1) $\frac{100mv}{t}$ (2) $\frac{200mv}{t}$

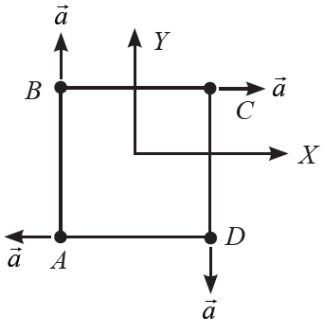
(3) 200 *mvt* (4) $\frac{mv}{100t}$

3. A force acts on a 2 kg object so that its position is given as a function of time as $x = 3t^2 + 5$. What is the work done by this force in first 5 seconds?

(1) 850 J (2) 950 J

(3) 875 J (4) 900 J

4. Four particles *A*, *B*, *C* and *D* with masses $m_A = m$, $m_B = 2m$, $m_C = 3m$ and $m_D = 4m$ are at the corners of a square. They have accelerations of equal magnitude with directions as shown. The acceleration of the centre of mass of the particles is:



(1) $\frac{a}{5}\left(\hat{i}-\hat{j}\right)$ (2) Zero

(3) $\frac{a}{5}\left(\hat{i}+\hat{j}\right)$ (4) $a\left(\hat{i}+\hat{j}\right)$

5. The radius of gyration of a uniform rod of length ℓ, about an axis passing through a point $\frac{\ell}{4}$ away from the centre of the rod and perpendicular to it is

(1) $\sqrt{\frac{7}{42}}\ell$

(2) $\sqrt{\frac{7}{48}}\ell$

(3) $\sqrt{\frac{5}{42}}\ell$

(4) $\sqrt{\frac{19}{48}}\ell$

6. Consider a spherical gaseous cloud of mass density $\rho(r)$ in free space where *r* is the radial distance from its center. The gaseous cloud is made of particles of equal mass *m* moving in circular orbits about the common center with the same kinetic energy *K*. The force acting on the particles is their mutual gravitational force.

If $\rho(r)$ is constant in time, the particle number density $n(r) = \rho(r)/m$ is

[*G* is universal gravitational constant]

(1) $\frac{K}{6\pi r^2 m^2 G}$ (2) $\frac{K}{\pi r^2 m^2 G}$

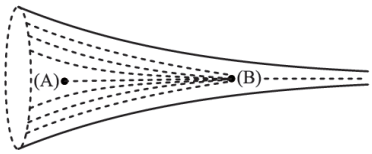
(3) $\frac{3K}{\pi r^2 m^2 G}$ (4) $\frac{K}{2\pi r^2 m^2 G}$

7. A massless wire of length *L* is hanging from a fixed support. The length changes to L_1 and L_2 when masses 1 kg and 2 kg are suspended respectively from its free end. Then the value of *L* is equal to:

(1) $\sqrt{L_1 L_2}$ (2) $\frac{L_1 + L_2}{2}$

(3) $2L_1 - L_2$ (4) $3L_1 - L_2$

8.



The figure shows a liquid of given density flowing steadily in horizontal tube of varying cross-section. Cross sectional areas at *A* is 1.5 cm^2 and *B* is 25 mm^2, if the speed of liquid at *B* is 60 cm/s then ($P_A - P_B$) is:

(Given P_A and P_B are liquid pressures at *A* and *B* points, Density ρ = 1000 kg m^{-3}, *A* and *B* are on the axis of tube)

(1) 175 Pa (2) 27 Pa

(3) 135 Pa (4) 36 Pa

9. Match Column-I with Column-II.

Column-I		Column-II	
A	3 Translational degrees of freedom	P	Monoatomic gases
B	3 Translational, 2 rotational degrees of freedoms	Q	Polyatomic gases
C	3 Translational, 2 rotational and 1 vibrational degrees of freedom	R	Rigid diatomic gases
D	3 Translational, 3 rotational and more than one vibrational degrees of freedom	S	Nonrigid diatomic gases

Choose the correct answer from the options given below:

(1) A → S; B → R; C → Q; D → P
(2) A → S; B → Q; C → P; D → R
(3) A → P; B → R; C → S; D → Q
(4) A → P; B → S; C → R; D → Q

10. Given below are two statements :

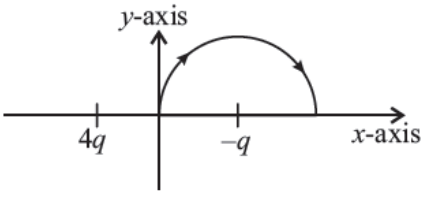
Statement-I: A second's pendulum has a time period of 1 second.

Statement-II: It takes precisely one second to move between the two extreme positions.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both Statement-I and Statement-II are false
(2) Statement-I is false but Statement-II is true
(3) Statement-I is true but Statement-II is false
(4) Both Statement-I and Statement-II are true

11. A two point charges $4q$ and $-q$ are fixed on the x-axis at $x = -d/2$ and $x = d/2$, respectively. If a third point charge 'q' is taken from the origin to $x = d$ along the semicircle as shown in the figure, the energy of the charge will



(1) Decrease by $\frac{q^2}{4\pi\varepsilon_0 d}$

(2) Decrease by $\frac{4q^2}{3\pi\varepsilon_0 d}$

(3) Increase by $\frac{2q^2}{3\pi\varepsilon_0 d}$

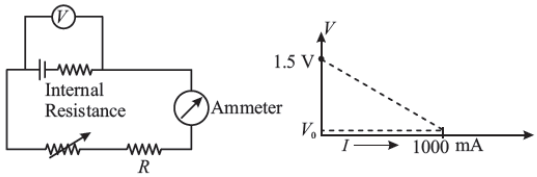
(4) Increase by $\frac{3q^2}{4\pi\varepsilon_0 d}$

12. Two identical thin metal plates has charge q_1 and q_2 respectively such that $q_1 > q_2$. The plates were brought close to each other to form a parallel plate capacitor of capacitance C. The potential difference between them is:

(1) $\frac{(q_1+q_2)}{C}$ (2) $\frac{(q_1-q_2)}{C}$

(3) $\left(\frac{q_1-q_2}{2C}\right)$ (4) $\frac{2(q_1-q_2)}{C}$

13. To verify Ohm's law, a student connects an ideal voltmeter across the battery as shown in the figure. The measured voltage is plotted as a function of the current, and the following graph is obtained. If V_0 is almost zero, identify the correct statement



(1) The emf of the battery is 1.5 V and its internal resistance is 1 Ω
(2) The emf of the battery is 3 V and its internal resistance is 1.5 Ω
(3) The emf of the battery is 1.5 V and its internal resistance is 1.5 Ω
(4) The potential difference across the battery is 1.5 V when it sends a current of 1000 mA

14. A coil having N turns is would tightly in the form of a spiral with inner and outer radii 'a' and 'b' respectively. Find the magnetic field at centre, when a current I passes through coil:

(1) $\frac{\mu_0 IN}{8}\left[\frac{a+b}{a-b}\right]$

(2) $\frac{\mu_0 IN}{2(b-a)}\log_e\left(\frac{b}{a}\right)$

(3) $\frac{\mu_0 IN}{4(a-b)}\log_e\left(\frac{b+a}{a}\right)$

(4) $\frac{\mu_0 IN}{8}\left(\frac{a-b}{a+b}\right)$

15. A particle having the same charge as of electron moves in a circular path of radius 0.5 cm under the influence of a magnetic field of 0.5 T. If an electric field of 100 V/m makes it to move in a straight path, then the mass of the particle is (Given charge of electron = 1.6×10^{-19} C)

(1) 9.1×10^{-31} kg (2) 1.6×10^{-27} kg
(3) 1.6×10^{-19} kg (4) 2.0×10^{-24} kg

16. A monochromatic light wave with wavelength λ_1 and frequency v_1 in air enters another medium. If the angle of incidence and angle of refraction at the interface are 45° and 30° respectively, then the wavelength λ_2 and frequency v_2 of the refracted wave are:

(1) $\lambda_2 = \lambda_1, v_2 = \sqrt{2}v_1$

(2) $\lambda_2 = \frac{1}{\sqrt{2}}\lambda_1, v_2 = v_1$

(3) $\lambda_2 = \sqrt{2}\lambda_1, v_2 = v_1$

(4) $\lambda_2 = \lambda_1, v_2 = \frac{1}{\sqrt{2}}v_1$

17. In a Young's double slits experiment, the ratio of amplitude of light coming from slits is 2 : 1. The ratio of the maximum to minimum intensity in the interference pattern is :

(1) 9 : 4 (2) 9 : 1

(3) 2 : 1 (4) 25 : 9

18. The electric field of a plane electromagnetic wave is given by $\vec{E} = E_0 \frac{\hat{i} + \hat{j}}{\sqrt{2}} \cos(kz + \omega t)$. At t = 0, a positively charged particle is at the point $(x, y, z) = \left(0, 0, \frac{\pi}{k}\right)$. If its instantaneous velocity at t = 0 is $v_0\hat{k}$, the force acting on it due to the wave is

(1) zero

(2) parallel to $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$

(3) parallel to $\hat{k}$

(4) antiparallel to $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$

19. An excited He^+ ion emits two photons in succession, with wavelengths 108.5 nm and 30.4 nm, in making a transition to ground state. The quantum number n, corresponding to its initial excited state is (for photon of wavelength λ, energy $E = \frac{1240 \text{ eV}}{\lambda(\text{in nm})}$

(1) n = 4 (2) n = 6

(3) n = 5 (4) n = 7

20. Consider the nuclear fission $Ne^{20} \rightarrow 2He^4 + C^{12}$. Given that the binding energy / nucleon of Ne^{20}, He^4 and C^{12} are, respectively, 8.03 MeV, 7.07 MeV, and 7.86 MeV. Identify the correct statement.

(1) Energy of 12.4 MeV will be supplied

(2) Energy of 9.72 MeV energy will be released

(3) Energy of 3.6 MeV will be released

(4) Energy of 11.9 MeV has to be supplied

Integer Type Questions

21. A one metre long (both ends open) organ pipe is kept in a gas that has double the density of air at STP. Assuming the speed of sound in air at STP is 300 m/s, the frequency difference between the fundamental and second harmonic of this pipe is $x\sqrt{2}$ Hz. Find the value of x.

22. The 1000 small droplets of water each of radius r and charge Q, make a big drop of spherical shape. The potential of big drop is how many times the potential of one small droplet?

23. A 100 V AC source of frequency 500 Hz is connected to LCR circuit with L = 8.1 mH, C = 12.5 μF and R = 10 Ω, all connected in series. Find the potential (in volt) across the resistance.

24. A particle A of mass 'm' and charge 'q' is accelerated by a potential difference of 50 V Another particle B of mass '4m' and charge 'q' is accelerated by a potential difference of 2500 V. The ratio of de-Broglie wavelength $\frac{\lambda_A}{\lambda_B}$ is $x\sqrt{2}$. Find the value of x.

25. M grams of steam at 100°C is mixed with 200 g of ice at its melting point in a thermally insulated container. If it produces liquid water at 40°C [heat of vaporization of water is 540 cal/ g and heat of fusion of ice is 80 cal/g], the value of M (in g) is _____.

SECTION-II (CHEMISTRY)

Single Correct Type Questions

26. The rate constant for the reaction: $2N_2O_5 \rightarrow 4NO_2 + O_2$ is $6 \times 10^{-5}\ s^{-1}$. If the rate is 2.40×10^{-5} mol $L^{-1}s^{-1}$, then the concentration of N_2O_5 in mol L^{-1} is

(1) 0.8 (2) 0.4
(3) 0.2 (4) 0.1

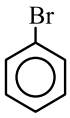
27. 25 mL of the given HCl solution requires 30 mL of 0.1 M sodium carbonate solution. What is the volume of this HCl solution required to titrate 30 mL of 0.2 M aqueous NaOH solution?

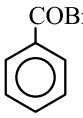
(1) 25 mL (2) 75 mL
(3) 50 mL (4) 12.5 mL

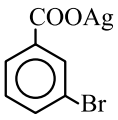
28. How many bonds are there in naphthalene?

(1) 18σ, 4π (2) 18 σ, 5π
(3) 19 σ, 5π (4) 19 σ, 4π

29. Silver benzoate reacts with bromine in acetone to form

(1) 

(2) 

(3) 
(4) 

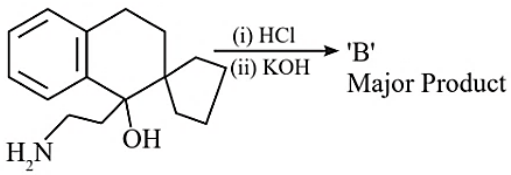
30. The most appropriate reagent for conversion of C_2H_5CN into $CH_3CH_2CH_2NH_2$ is:

(1) $NaBH_4$ (2) CaH_2
(3) $Na(CN)BH_3$ (4) $LiAlH_4$

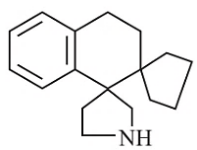
31. In general, the property (magnitudes only) that shows an opposite trend in comparison to other properties across a period is :

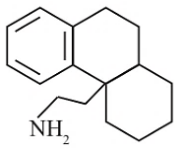
(1) Ionization enthalpy
(2) Electron gain enthalpy
(3) Atomic radius
(4) Electronegativity.

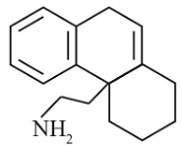
32. In the reaction given below:

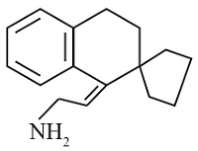


'B' is :

(1) 

(2) 

(3) 

(4) 

33. Match Column-I with Column -II:

Column-I (Chemicals)		Column-II (Use/Preparation/ Constituent)	
A	Alcoholic potassium hydroxide	P	Electrodes in batteries
B	$Pd/BaSO_4$	Q	Obtained by addition reaction
C	BHC (Benzene hexachloride)	R	Used for β-elimination reaction
D	Graphite	S	Lindlar's Catalyst

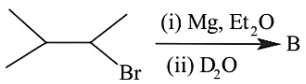
Choose the correct answer from the options given below:

(1) A→R; B→P; C→Q; D→S
(2) A→P; B→P; C→S; D→R
(3) A→P; B→Q; C→S; D→R
(4) A→R; B→S; C→Q; D→P

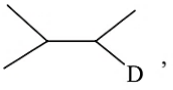
34. The factor that does not affect the percent covalent character of the ionic bond is____

(1) Polarising power of cation
(2) Extent of distortion of anion
(3) Polarisability of the anion
(4) Polarising power of anion

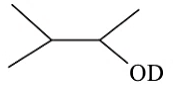
35. Compound 'A' undergoes following sequence of reactions to give compound 'B'. The correct structure and chirality of compound 'B' is [where Et is – C_2H_5]



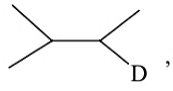
Compound 'A'

(1) 

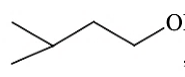
, Achiral

(2) 

, chiral

(3) 

, chiral

(4) 

, Achiral

36. Identify the incorrect statement from the following:

(1) Amylose is a branched chain polymer of glucose

(2) Starch is a polymer of α-D-glucose

(3) β-Glycosidic linkage makes cellulose polymer

(4) Glycogen is called as animal starch

37. 5 moles of AB_2 weigh 125×10^{-3} kg and 10 moles of A_2B_2 weigh 300×10^{-3} kg. The molar mass of $A_{(M_A)}$ and molar mass of $B_{(M_B)}$ in kg mol^{-1} are:

(1) $M_A = 50 \times 10^{-3}$ and $M_B = 25 \times 10^{-3}$

(2) $M_A = 25 \times 10^{-3}$ and $M_B = 50 \times 10^{-3}$

(3) $M_A = 5 \times 10^{-3}$ and $M_B = 10 \times 10^{-3}$

(4) $M_A = 10 \times 10^{-3}$ and $M_B = 5 \times 10^{-3}$

38. The correct order of the spin-only magnetic moment of metal ions in the following low spin complexes,

$[V(CN)_6]^{4-}, [Fe(CN)_6]^{4-}, [Ru(NH_3)_6]^{3+}$ and $[Cr(NH_3)_6]^{2+}$ is:

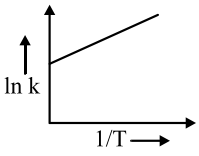
(1) $V^{2+} > Cr^{2+} > Ru^{3+} > Fe^{2+}$

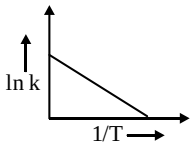
(2) $V^{2+} > Ru^{3+} > Cr^{2+} > Fe^{2+}$

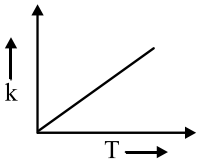
(3) $Cr^{2+} > V^{2+} > Ru^{3+} > Fe^{2+}$

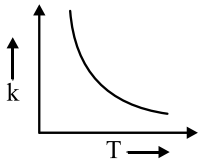
(4) $Cr^{2+} > Ru^{3+} > Fe^{2+} > V^{2+}$

39. Which one of the following given graphs represents the variation of rate constant (k) with temperature (T) for an endothermic reaction?

(1) 

(2) 

(3) 

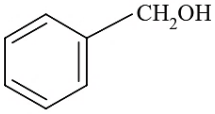
(4) 

40. Which one of the following compounds does not exhibit resonance?

(1) $CH_3CH_2CH_2CONH_2$

(2) $CH_3CH_2CH = CHCH_2NH_2$

(3) $CH_3CH_2OCH = CH_2$

(4) 

41. Given below are two statements: one is labelled as

Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Cu^{2+} in water is more stable than Cu^{+}.

Reason (R): Enthalpy of hydration for Cu^{2+} is much more negative than that of Cu^{+}.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

42. The pH of a 0.02 M NH_4C1 solution will be (given: $K_b(NH_4OH) = 10^{-5}$ and log 2= 0.301)

(1) 4.65

(2) 5.35

(3) 4.35

(4) 2.65

43. 29.5 mg of an organic compound containing nitrogen was digested according to Kjeldahl's method and the evolved ammonia was absorbed in 20 mL of 0.1 M HCl solution. The excess of the acid required 15 mL of 0.1 M NaOH solution for complete neutralization. The percentage of nitrogen in the compound is

(1) 59.0

(2) 47.4

(3) 23.7

(4) 29.5

44. Which of the following is correct relation between mole fraction of solute (X_2) and molarity of solution (M) ?

Here, X_1 is mole fraction of solvent, M_1 and M_2 are molar mass of solvent and solute respectively in g/mol. ρ is density of solution in g/L.

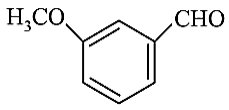
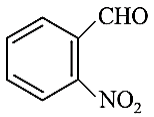
(1) $M = \dfrac{X_2}{M_2(X_1M_1 + x_2M_2)}$

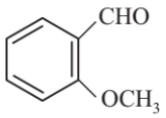
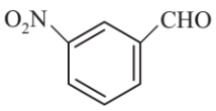
(2) $M = \dfrac{(x_1M_1 + x_2M_2)}{x_2M_2}$

(3) $M = \dfrac{x_2\rho}{(x_1M_1 + x_2M_2)}$

(4) $M = \dfrac{(1 - m\rho)}{mM_2}$

45. The increasing order of the following compounds towards HCN addition is

(*i*)  (*ii*) 

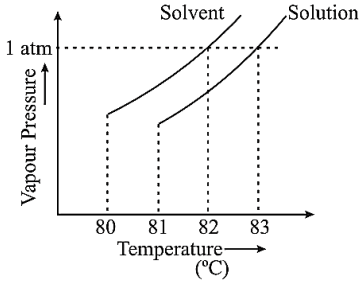
(*iii*)  (*iv*) 

(1) (iv) < (i) < (ii) < (iii)
(2) (iii) < (i) < (iv) < (ii)
(3) (ii) < (iv) < (i) < (iii)
(4) (iv) < (i) < (iii) < (ii)

Integer Type Questions

46. The orbital angular momentum of an electron in 3s orbital is $\frac{xh}{2\pi}$. The value of x is

47. The vapour pressure vs. temperature curve for a solution solvent system is shown below.



The boiling point of the solvent is _____°C.

48. A KCl solution of conductivity 0.14 S m^{-1} shows a resistance of 4.19Ω in a conductivity cell. If the same cell is filled with an HCl solution, the resistance drops to 1.03Ω. The conductivity of the HCl solution is _____× 10^{-2} Sm^{-1}. (Round off to the nearest integer).

49. The number of species from the following carrying a single lone pair on central atom Xenon is ____:

$XeF_5^+, XeO_3, XeO_2F_2, XeF_5^-, XeO_3F_2, XeOF_4, XeF_4$

50. The enthalpy of combustion of propane, graphite and dihydrogen at 298 K are –2220.0 kJ mol^{-1}, –393.5kJ mol^{-1} and –285.8kJ mol^{-1} respectively. The magnitude of the enthalpy of formation of propane C_3H_8 is _____ kJ mol^{-1} (Nearest integer).

SECTION-III (MATHEMATICS)

Single Correct Type Questions

51. The number of real roots of the equation, $e^{4x}+e^{3x}-4e^{2x}+e^{x}+1=0$

(1) 3 (2) 1
(3) 4 (4) 2

52. Let $B=\begin{bmatrix}1 & 3 & \alpha\\ 1 & 2 & 3\\ \alpha & \alpha & 4\end{bmatrix}, \alpha>2$ be the adjoint of a matrix A and |*A*| = 2, then $\begin{bmatrix}\alpha & -2\alpha & \alpha\end{bmatrix}B\begin{bmatrix}\alpha\\ -2\alpha\\ \alpha\end{bmatrix}$ is equal to:

(1) 16 (2) 32
(3) –16 (4) 0

53. The coefficient of x^4 in the expansion of $(1 + x + x^2 + x^3)^6$ in powers of *x*, is _____.

(1) 120 (2) 220
(3) 320 (4) 100

54. The complex number $z=\frac{i-1}{\cos\frac{\pi}{3}+i\sin\frac{\pi}{3}}$ is equal to:

(1) $\sqrt{2}\left(\cos\frac{5\pi}{12}+i\sin\frac{5\pi}{12}\right)$

(2) $\cos\frac{\pi}{12}-i\sin\frac{\pi}{12}$

(3) $\sqrt{2}\left(\cos\frac{\pi}{12}+i\sin\frac{\pi}{12}\right)$

(4) $\sqrt{2}i\left(\cos\frac{5\pi}{12}-i\sin\frac{5\pi}{12}\right)$

55. A point *P* moves so that the sum of squares of its distances from the points (1,2) and (—2,1) is 14. Let *f*(*x*, *y*) = 0 be the locus of *P*, which intersects the *x*-axis at the points *A*, *B* and the y-axis at the points *C*, *D*. Then the area of the quadrilateral *ACBD* is equal to:

(1) $\frac{9}{2}$ (2) $\frac{3\sqrt{17}}{2}$
(3) $\frac{3\sqrt{17}}{4}$ (4) 9

56. The foot of perpendicular of the point (2, 0, 5) on the line $\frac{x+1}{2}=\frac{y-1}{5}=\frac{z+1}{-1}$ is (α,β,γ). Then.

Which of the following is NOT correct?

(1) $\frac{\alpha\beta}{\gamma}=\frac{4}{15}$

(2) $\frac{\alpha}{\beta}=-8$

(3) $\frac{\beta}{\gamma}=-5$

(4) $\frac{\gamma}{\alpha}=\frac{5}{8}$

57. Let μ be the mean and σ be the standard deviation of the distribution

x_1	0	1	2	3	4	5
f_1	$k+2$	$2k$	k^2-1	k^2-1	k^2+1	$k-3$

Where $\sum f_i = 62$. If [*x*] denotes the greatest integer $\le x$, then $\left[\mu^2+\sigma^2\right]$ is equal

(1) 8 (2) 7

(3) 6 (4) 9

58. If the sum of the first 40 terms of the series, 3 + 4 + 8 + 9 + 13 + 14 + 18 + 19 + ... is (102)*m*, then *m* is equal to:

(1) 25 (2) 20

(3) 10 (4) 5

59. A circle of radius 2 units passes through the vertex and the forcus of the parbola y^2 = 2x and touches the parabola $y=\left(x-\frac{1}{4}\right)^2+\alpha$, where $\alpha>0$. Then $(4\alpha-8)^2$ is equal to _____.

(1) 73 (2) 63

(3) 89 (4) 20

60. If [*t*] denotes the greatest integer $\le$ *t*, then number of points at which the function

$$f(x)=4|2x+3|+9\left[x+\frac{1}{2}\right]-12[x+20]$$

Is not differentiable in the open interval (–20, 20), is_____.

(1) 79 (2) 80

(3) 40 (4) 21

61. Number of 4-digit numbers (the repetition of digits is allowed) which are made using the digits 1,2,3 and 5, and are divisible by 15, is equal to

(1) 21 (2) 16

(3) 22 (4) 15

62. $\operatorname{cosec}\left[2\cot^{-1}(5)+\cos^{-1}\left(\frac{4}{5}\right)\right]$ is equal to:

(1) $\frac{65}{33}$ (2) $\frac{56}{33}$

(3) $\frac{65}{56}$ (4) $\frac{75}{56}$

63. Let *A* = {0, 3, 4, 6, 7, 8, 9, 10} and *R* be the relation defined on *A* such that *R* = {(*x*, *y*) ∈ *A* × *A*: *x* – *y* is odd positive integer or *x* – *y* = 2}. The minimum number of elements that must be added to the relation *R*, so that it is a symmetric relation, is equal to

(1) 20 (2) 13

(3) 7 (4) 19

64. Considering only the principal values of inverse function, the set

$$A=\left\{x\ge 0:\tan^{-1}(2x)+\tan^{-1}(3x)=\frac{\pi}{4}\right\}$$

(1) Contains two elements

(2) Contains more than two elements

(3) Is a singleton

(4) Is an empty set

65. Let $\vec{a}=2\hat{i}-7\hat{j}+5\hat{k},\vec{b}=\hat{i}+\hat{k}$ and $\vec{c}=\hat{i}+2\hat{j}-3\hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r}\times\vec{a}=\vec{c}\times\vec{a}$ and $\vec{r}\cdot\vec{b}=0$, then $|\vec{r}|$ is equal to:

(1) $\frac{11}{7}\sqrt{2}$ (2) $\frac{11}{7}$

(3) $\frac{11}{5}\sqrt{2}$ (4) $\frac{\sqrt{914}}{7}$

66. Let be a hyperbola such that the sum of length s of the transverse and the conjugate axes is $4(2\sqrt{2}+\sqrt{14})$. If the eccentricity *H* is $\frac{\sqrt{11}}{2}$, the the value of $a^2 + b^2$ is equal to.

(1) 66 (2) 88

(3) 22 (4) 33

67. Let $\frac{dy}{dx}=\frac{ax-by+a}{bx+cy+a}$, where *a*, b, *c*, are constants, represent a circle passing through the point (2, 5). Then the shortest distance of the point (11,6) from this circle is:

(1) 10

(2) 8

(3) 7

(4) 5

68. If the area of the bounded region $R=\left\{(x,y):\max\{0,\log_e x\}\le y\le 2^x,\frac{1}{2}\le x\le 2\right\}$ is, $\alpha(\log_e 2)^{-1}+\beta(\log_e 2)+\gamma$ then the value of $(\alpha+\beta-2\gamma)^2$ is equal to:

(1) 8
(2) 2
(3) 4
(4) 1

69. Let $f:R\to R$ be given by $f(x)=(x-1)(x-2)(x-5)$.

Define $F(x)=\int_0^x f(t)dt, x>0$. Then which of the following option is incorrect?

(1) F has a local minimum at $x = 1$
(2) F has a local maximum at $x = 2$
(3) $F(x)\neq 0$ for all $x\in(0,5)$
(4) F has two local maxima and one local minimum in (0, ∞)

70. The remainder on dividing 5^{99} by 11 is _____.

(1) 9
(2) 6
(3) 7
(4) 4

Integer Type Questions

71. The value of b > 3 for which $12\int_3^b \frac{1}{(x^2-1)(x^2-4)}dx=\log_e\left(\frac{49}{40}\right)$ is equal to_____.

72. Let

$A=\{(x,y)\in R\times R|\ 2x^2+2y^2-2x-2y=1\}$

$B=\{(x,y)\in R\times R|\ 4x^2+4y^2-16y+7=0\}$ and

$C=\{(x,y)\in R\times R|\ x^2+y^2-4x-2y+5\le r^2\}$.

Then the minimum value of $2|r|-2\sqrt{5}$ such that $A\cup B\subseteq C$ is equal to:

73. If the integral $\int\frac{1}{\sqrt[4]{(x-1)^3(x+2)^5}}dx$ $\frac{a}{9}\left(\frac{x-1}{x+2}\right)^{1/4}+C$ then a is equal (where C is a constant of integration)

74. The coefficients a, b and c of the quadratic equation, $ax^2 + bx + c = 0$ are obtained by throwing a dice three times. If the probability that this equation has equal roots is P, then the value of $\frac{432}{5}P$ is

75. If $\cos^2 10° - \cos10° \cos50° + \cos^2 50°$ is l then $\frac{4l}{3}$ is



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if